



Forklift Operational Safety

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What is a forklift?

The forklift is a powerful instrument, which helps a person to lift and transfer heavy and large loads accurately with less effort. Using equipment such as forklifts, carts, or hand trucks, lifting and transporting loads instead of carrying loads reduces the risk of back injuries.





Forklift Accident Statistics

- Annual costs: about \$135 million in direct and indirect costs
- Accident Rate: About 11% of forklifts are involved in accidents per year
- Number of injuries: 61,800+ non-serious and 34,900 seriously injured per year

The most common accidents:

- Forklift overturned – 22%
- Pushing a pedestrian – 20%
- Crushed under the forklift – 16%
- Falling from the forklift – 9%



গাড়ি বনাম ফর্কলিফট: প্রধান পার্থক্য

বৈশিষ্ট্য	গাড়ি (Car)	ফর্কলিফট (Forklift)
ওজন	২,০০০ - ৬,০০০ পাউন্ড	৪,০০০ - ৪০,০০০ পাউন্ড
স্টিয়ারিং ব্যবস্থা	সামনের চাকায় স্টিয়ারিং	পিছনের চাকায় স্টিয়ারিং
ভার কেন্দ্র (Center of Gravity)	নিম্ন (Low Center of Gravity)	উচ্চ (High Center of Gravity)
দৃশ্যমানতা	চমৎকার দৃশ্যমানতা (Excellent Visibility)	কম দৃশ্যমানতা (Low Visibility)



Common Causes of Accidents

- Driving with high forks.
- Rotating or braking sharp angles with loads.
- Carrying unstable loads.
- Collision with overhead structure.





Cars vs Forklifts: Main Differences





Forklift stability triangle

The "stability triangle" is an important concept for balancing the forklift. It is an imaginary triangle formed by two front wheels and rear axle pivot points of the forklift.

The main concept of the stability triangle:

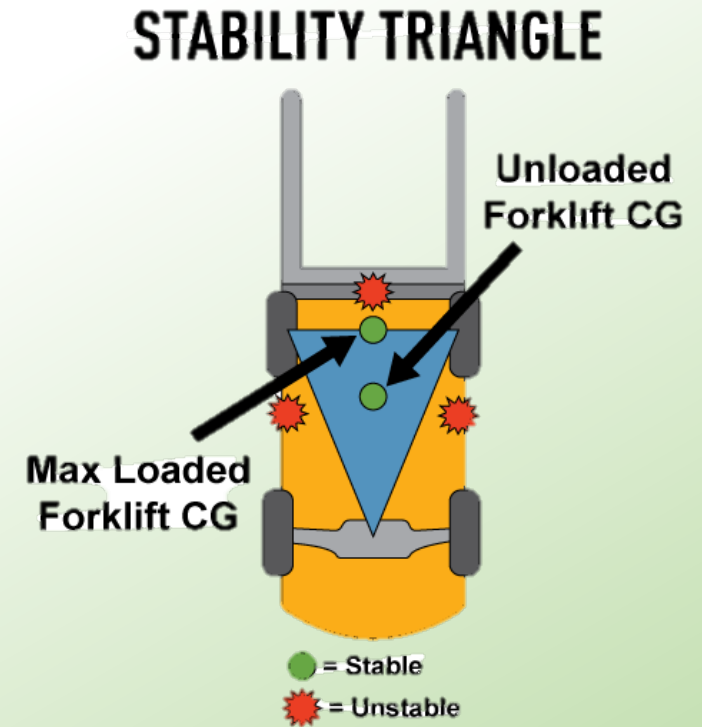
The three points of the triangle:

- Front two wheels (main point of load bearing)
- Rear Pivot Point (where the steering is controlled)

Balance Keeping:

If the load of the forklift is within the triangle, then the forklift will be stable.

If the load moves out of the triangle, the forklift may overturn.





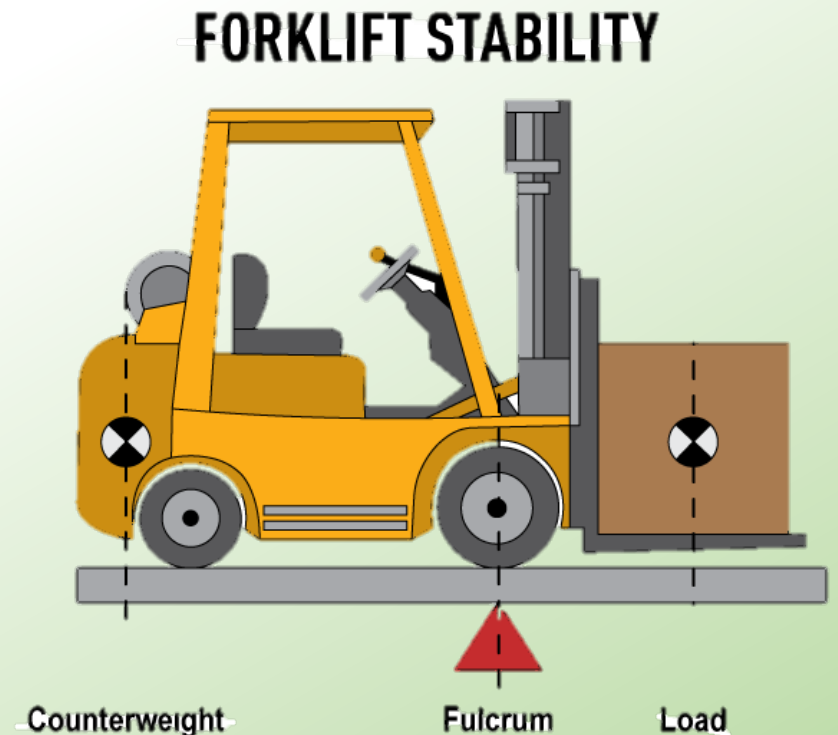
Forklift stability triangle

Risks & Prevention::

- Excessive or unbalanced load may cause the forklift to overturn !
- Making a quick turn or moving on a slope can lead to loss of balance.
- The higher the load, the lower the stability.

Safe handling tips:

- Always keep the load at a low height !
- Take a slow and step-by-step turn.
- Avoid overloading and balance the load.





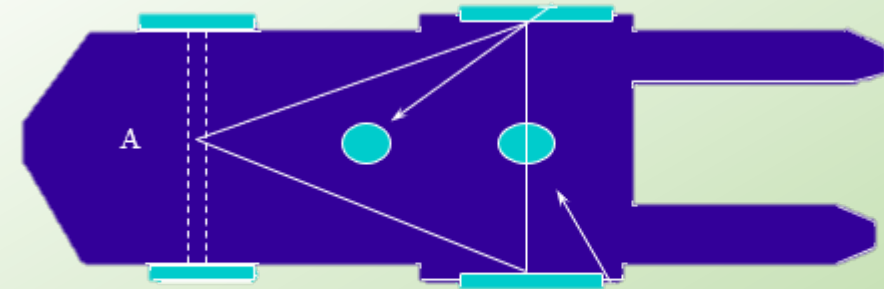
Forklift stability triangle

If the forklift is loaded:

- The combined load center (CG) moves forward, as the load is in front of the forklift.
- However, in reality, the load center should never reach the front wheel, as this will make the forklift very unsafe and increase the risk of overturning in the front.

Effects of extra counterweights:

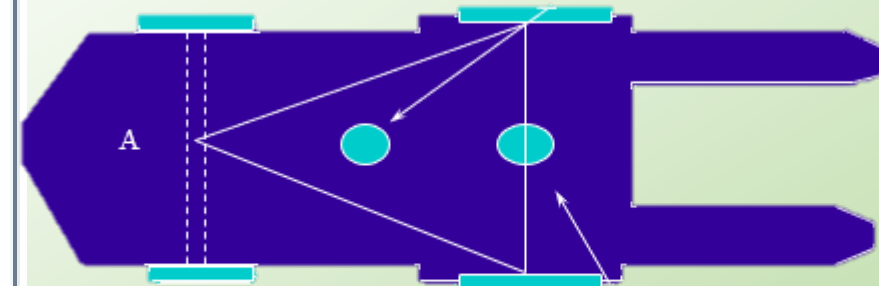
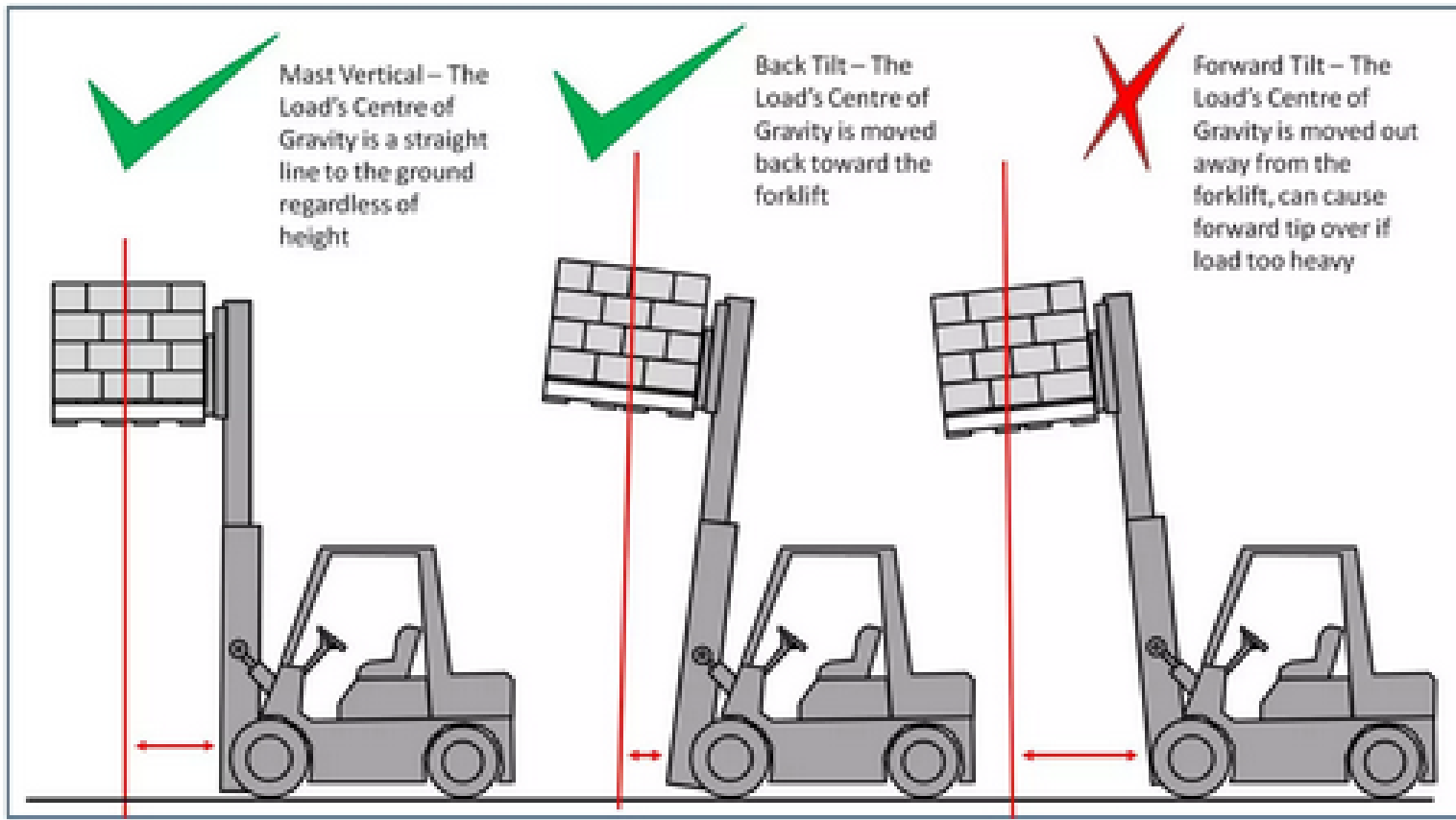
- Adding additional counterweight to the forklift, its center of charge will move towards point A (behind the stability triangle).
- This can improve the forward balance, but also reduce the side-to-side balance.
- If the load center of the forklift moves too backwards, it increases the likelihood of overturning laterally (sideways), especially when making sharp turns.





Load Center

The load center is the distance from the starting point of the forklift to the center of the load, which is usually measured in inches or millimeters.





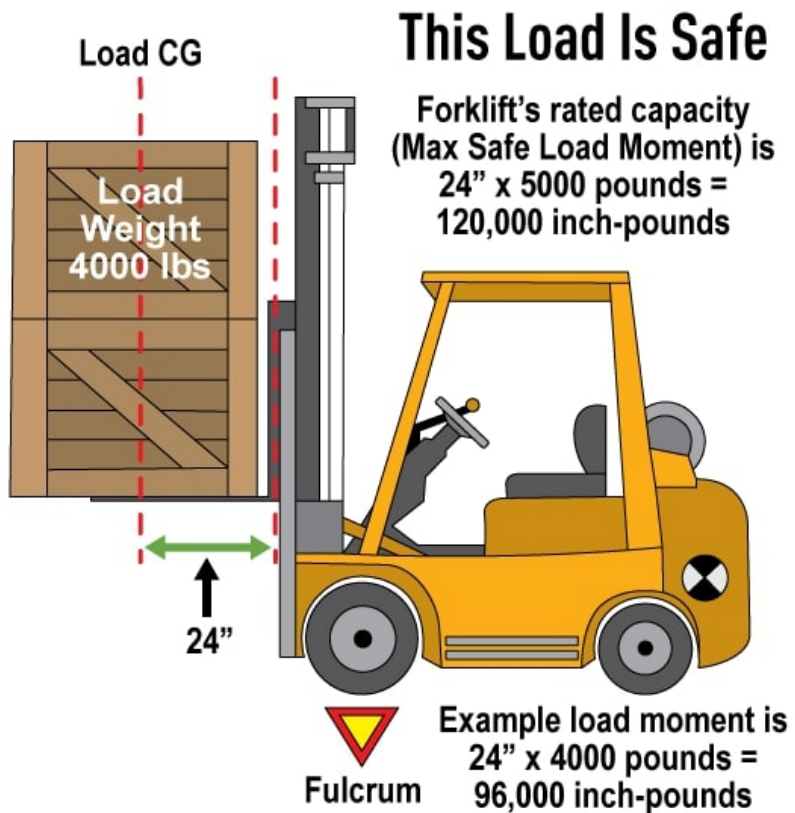
Capacity Plate

TOYOTA ELECTRIC FORKLIFT TRUCK									
MODEL	8FBCU25-COMP				SERIAL NO.	82628			
MAST	QFV	BACK TILT	5		ATTACH	SIDE SHIFTER			
TYPE	E	VOLTAGE	36	V	BATTERY TYPE	E O	MAX AMPERE HOUR CAPACITY	1210	AH
FRONT TREAD	39.4 in	TIRE FR	21X7X15/SOLID						
	1000 mm	SIZE RR	16X5X10-1/2/SOLID						
TRUCK WEIGHT W/O BATTERY	8990 lb		BATTERY WEIGHT		2400 lb/ 3154 lb				
ACCURACY ±5%	4080 kg		MIN/MAX		1090 kg/ 1430 kg				
RATED CAPACITY WITH VERTICAL MAST EQUIPPED AT MAX. LIFT HEIGHT 'C' AS SHOWN.									
						CAPACITY			
		A		B		C		D	
		in	24	24	278	0	2330	lb	THIS FORKLIFT TRUCK MEETS OR EXCEEDS DESIGN SPECIFICATIONS OF ANSI/ITSDF B56.1 IN EFFECT ON THE DATE OF MANUFACTURE.
		mm	600	600	7015	0	1060	kg	
		A		B		C		D	
		in	30	30	278	0	2060	lb	
		A		B		C		D	
		mm	760	760	7015	0	920	kg	
WARNING IMPROPER OPERATION OR MAINTENANCE COULD RESULT IN INJURY OR DEATH. TRAINED OPERATORS ONLY. READ OPERATOR'S MANUAL FIRST. 57899-U2260-71									

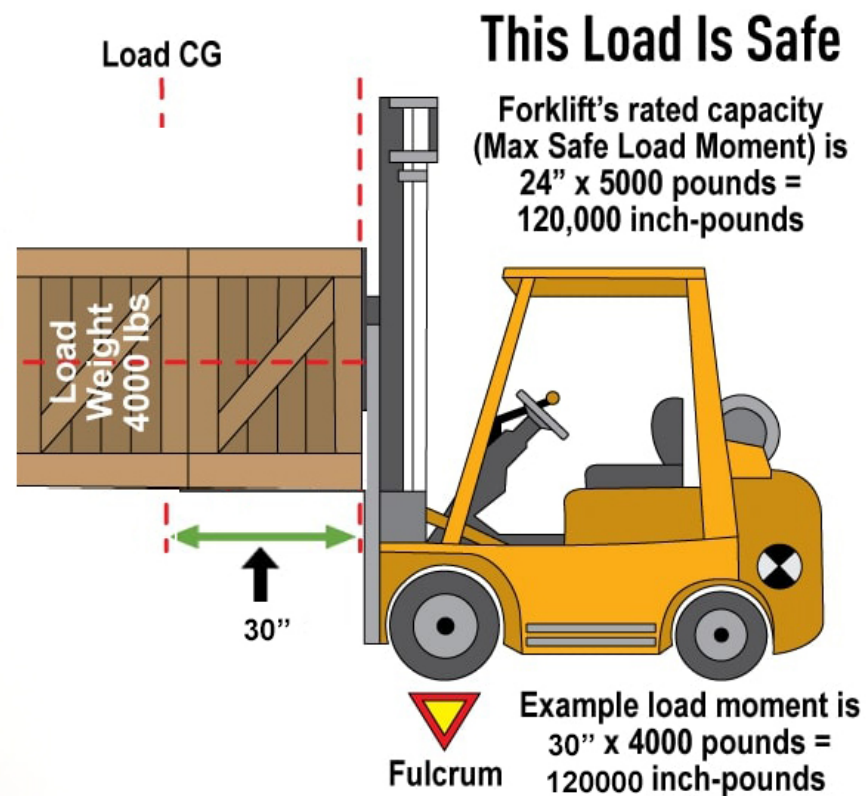
C2 ENGLISH 57647-4 1004-71



Safe load limits



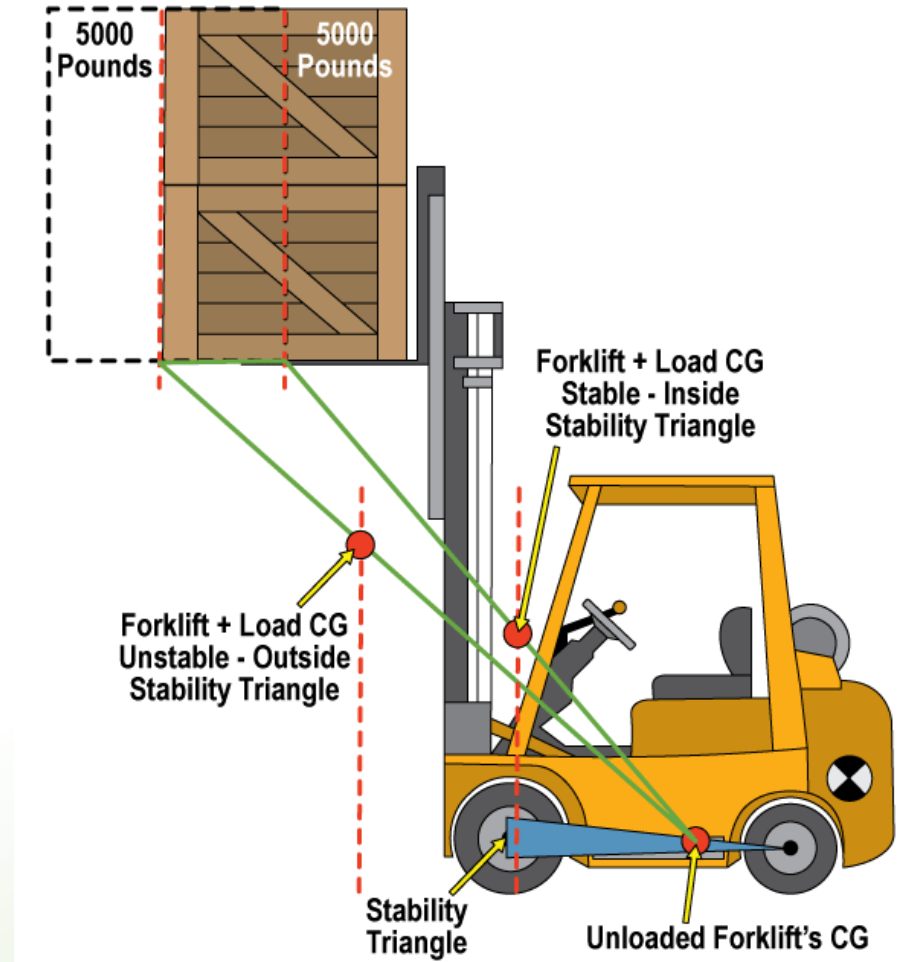
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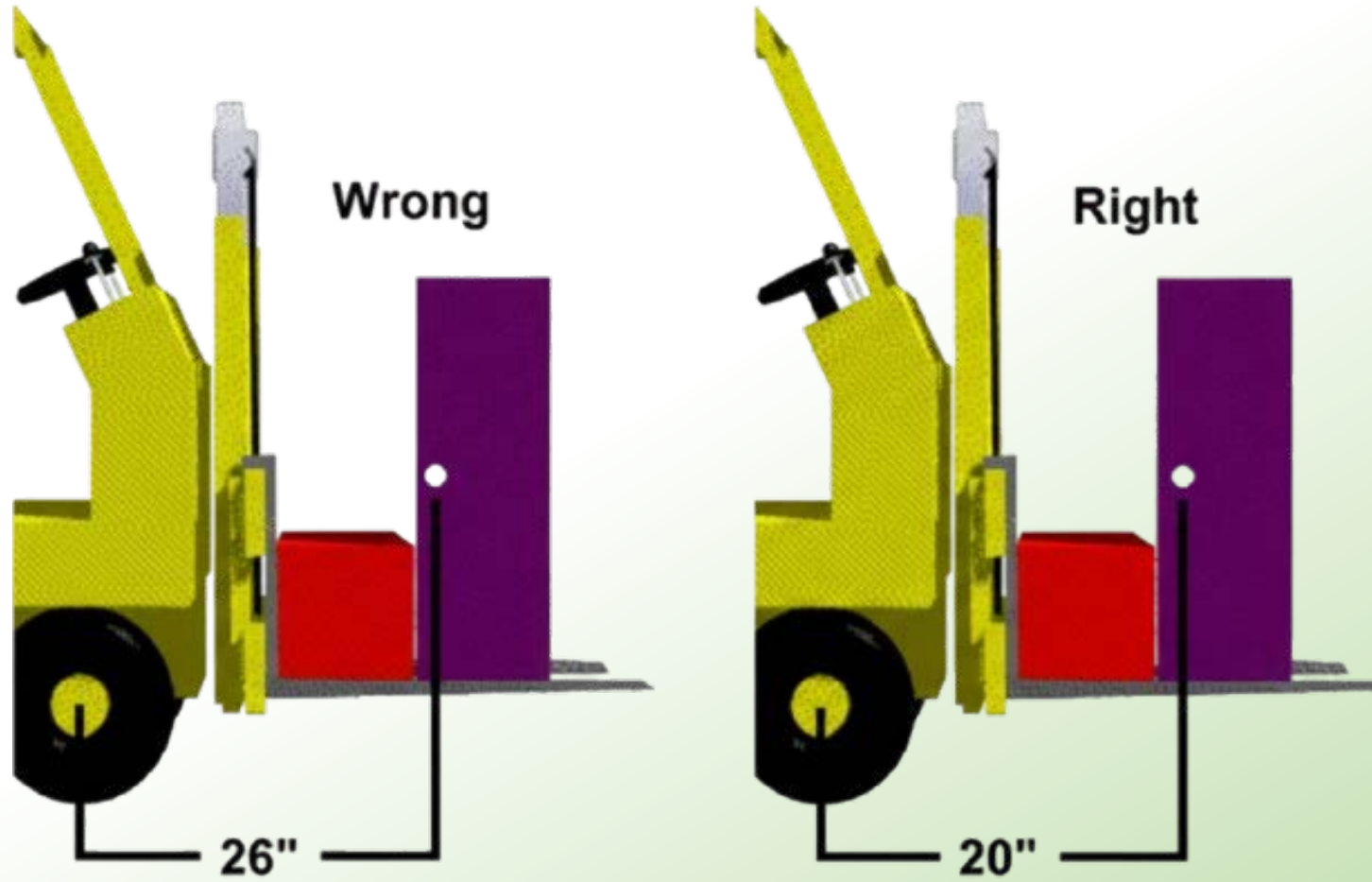


Safe load limits



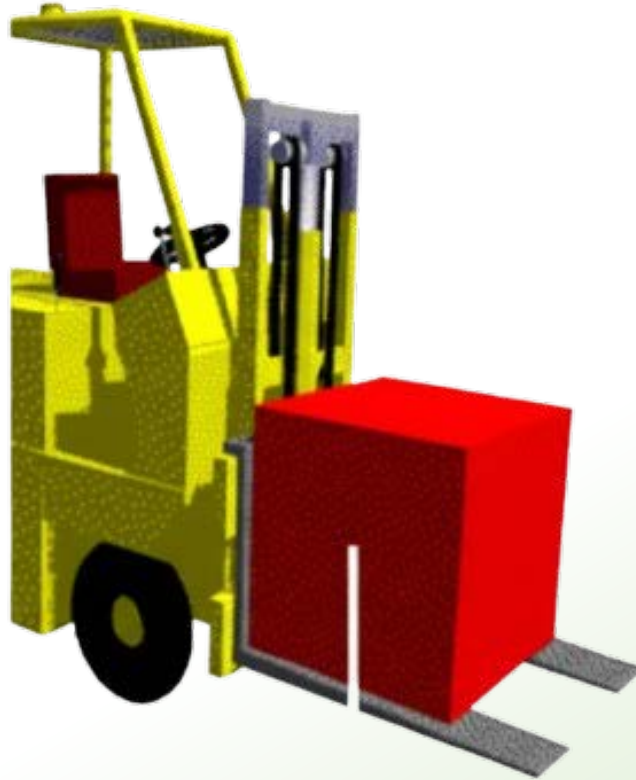


Safe load limits





Safe load limits



24 inch



22 inch



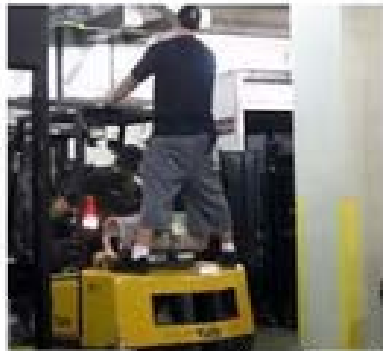
Unsafe forklift handling

The combined weight center stability of the forklift and load can go beyond the stability triangle if: :

- The load is lifted to the tip of the fork,
- The load leans more forward,
- The load is extra wide,
- The movement of the forklift causes the load center to shift.



Unsafe forklift handling





Handling and Moving Loads

Pick-up a Load

- Approach the load slowly and straight on (Load centered)
- Stop when the tips of the forks are about a foot away.
- Level forks and adjust fork height
- Move forward slowly until load is fully against backrest
- Lift the load high enough to clear floor obstructions
- Back the load out slightly from the storage location
- Carefully tilt mast back to stabilize the load
- Ensure rear is clear; move backwards



Handling and Moving Loads

Put down a Load

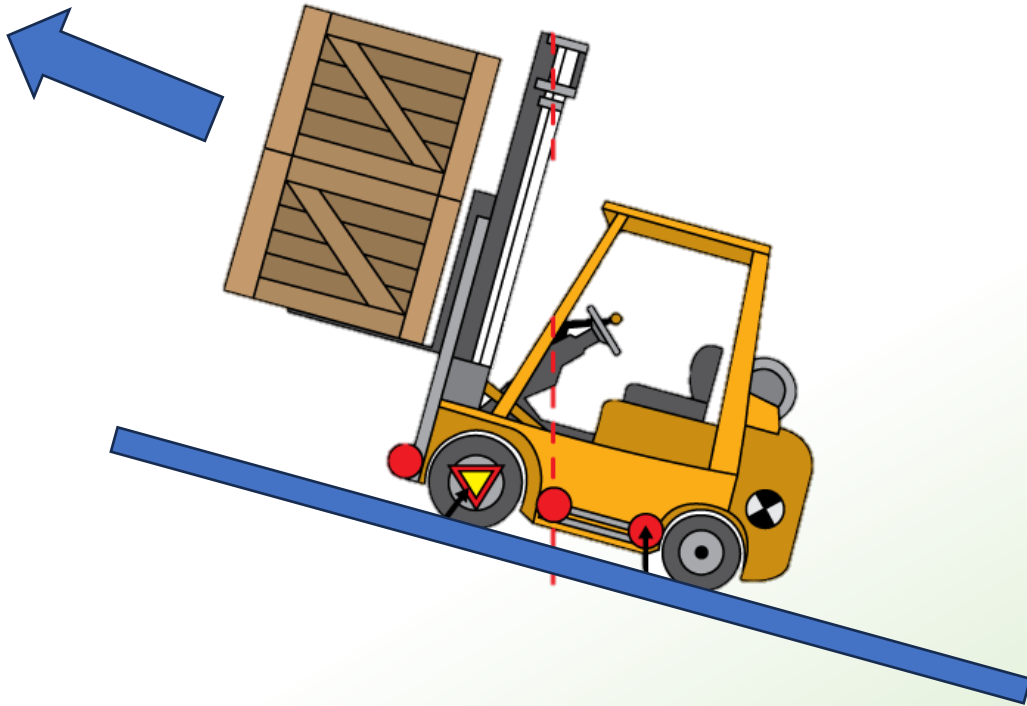
- Approach the unloading point straight on
- Stop when the tips of the forks or side of the load are about a foot away
- Lower load, carefully level forks
- Move forward slowly until load is in desired location
- Lower the load to floor or storage location
- Ensure forks are clear of pallet
- Ensure rear is clear; move backwards



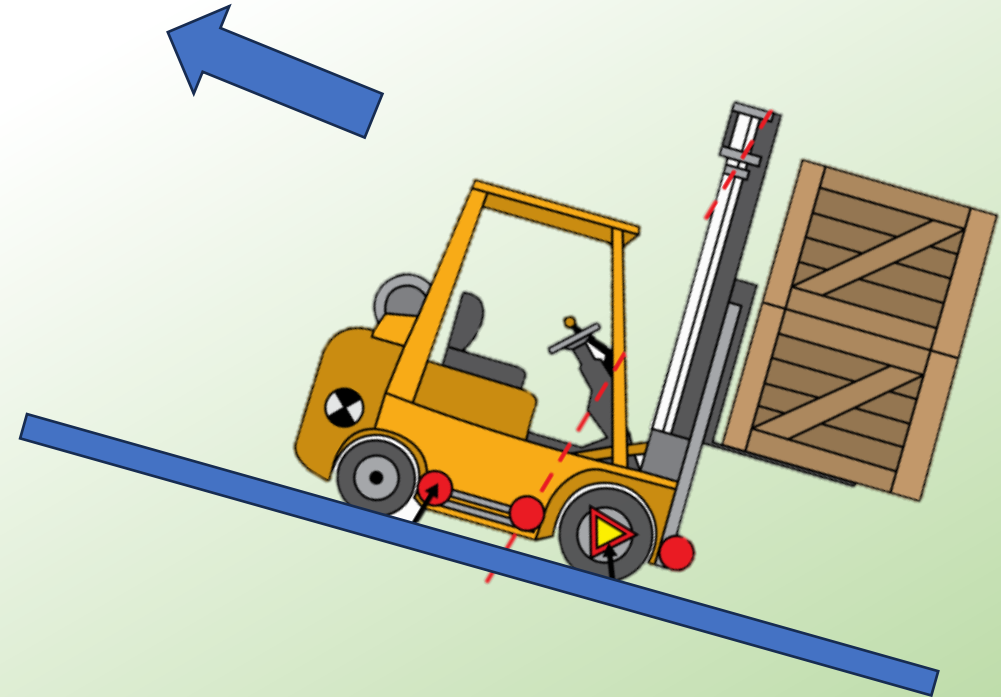
Secure Forklift Operation

What is the right method when climbing up with a load?

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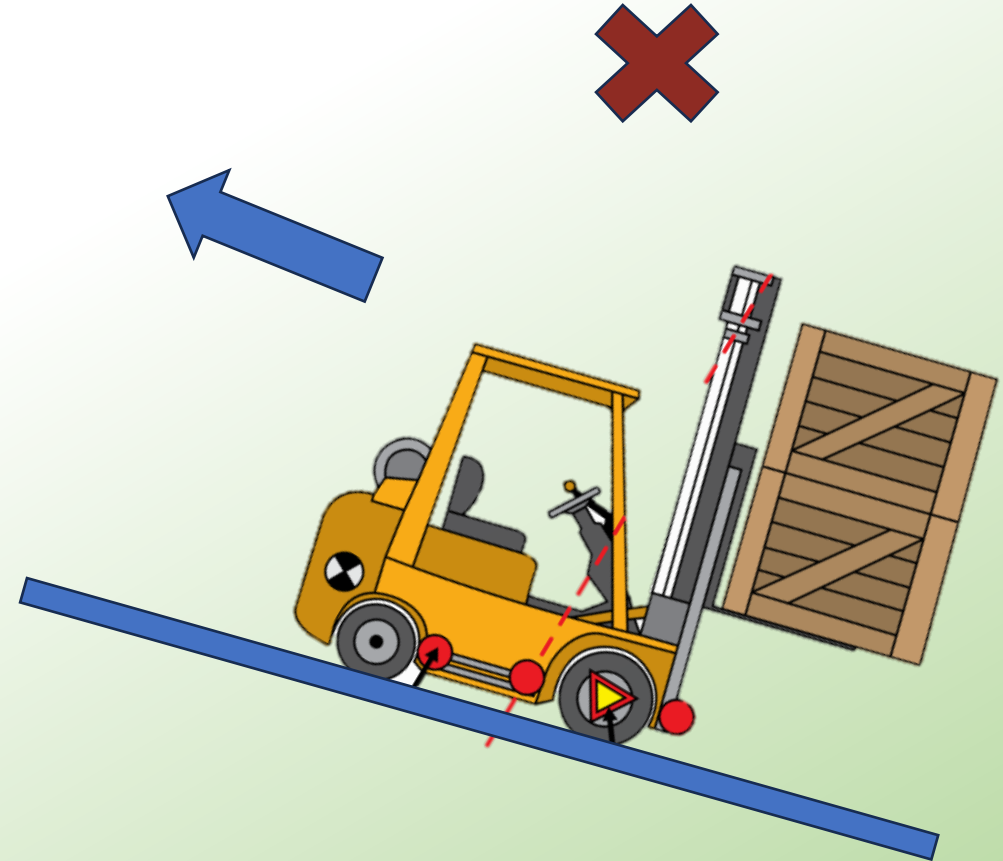
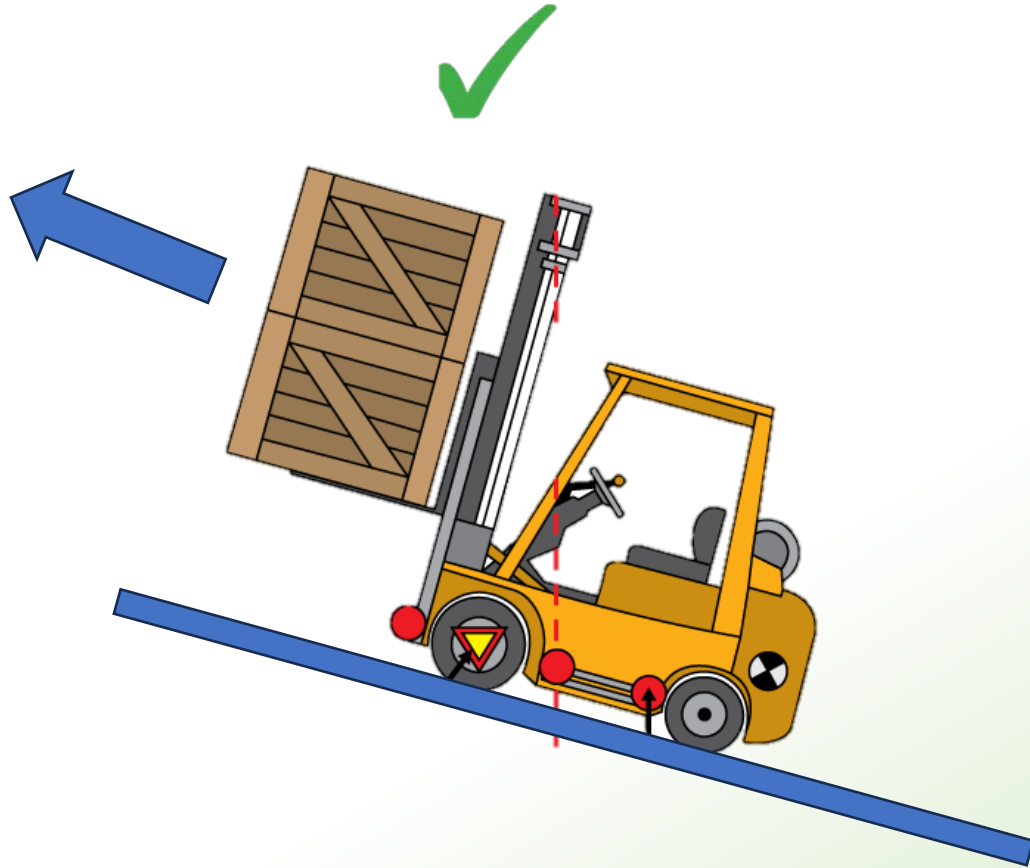
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Secure Forklift Operation

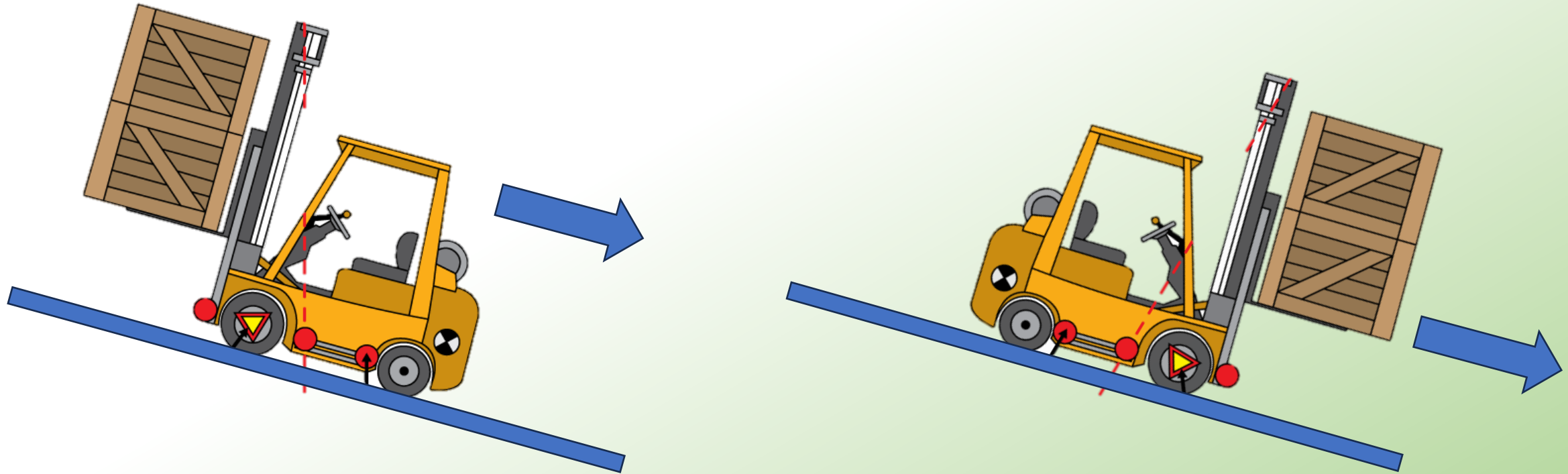
The correct method when climbing up with the load





Secure Forklift Operation

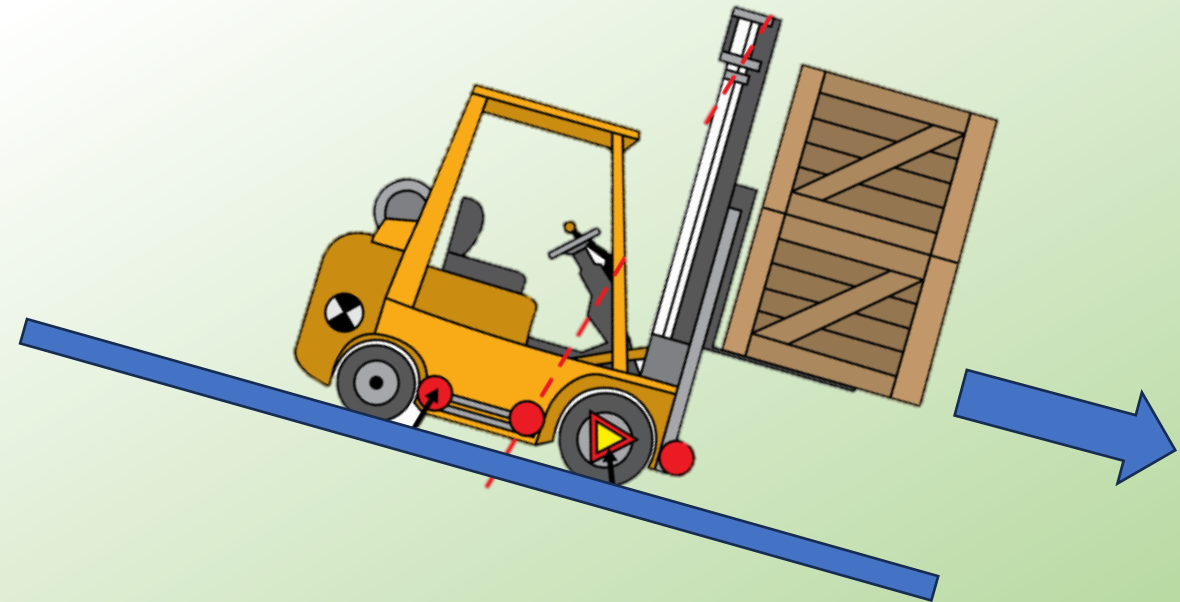
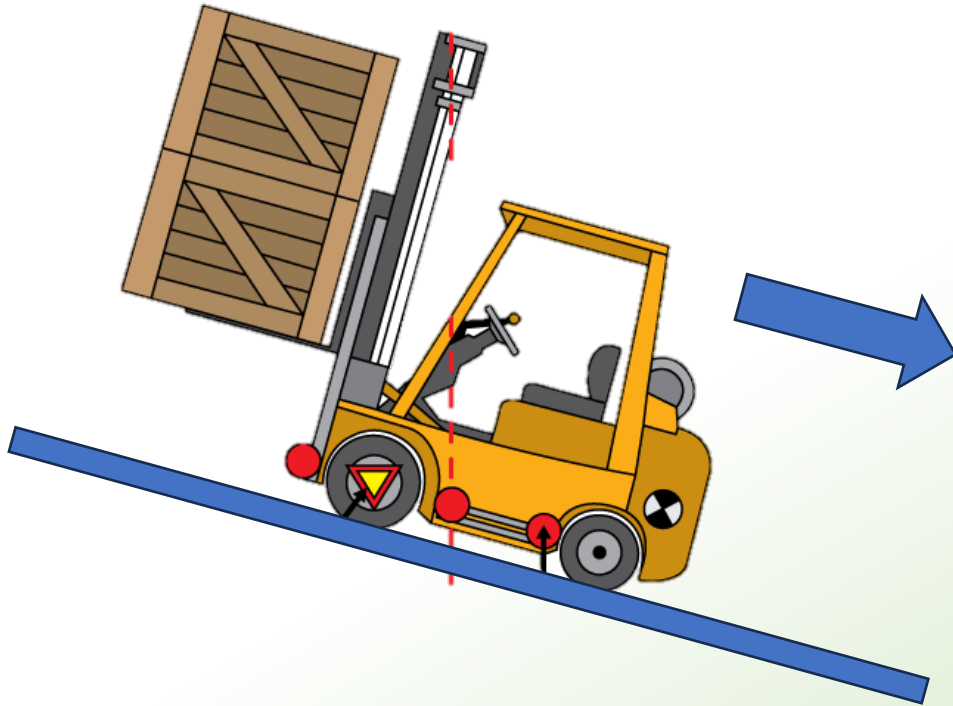
What is the correct method when going down with a load?





Secure Forklift Operation

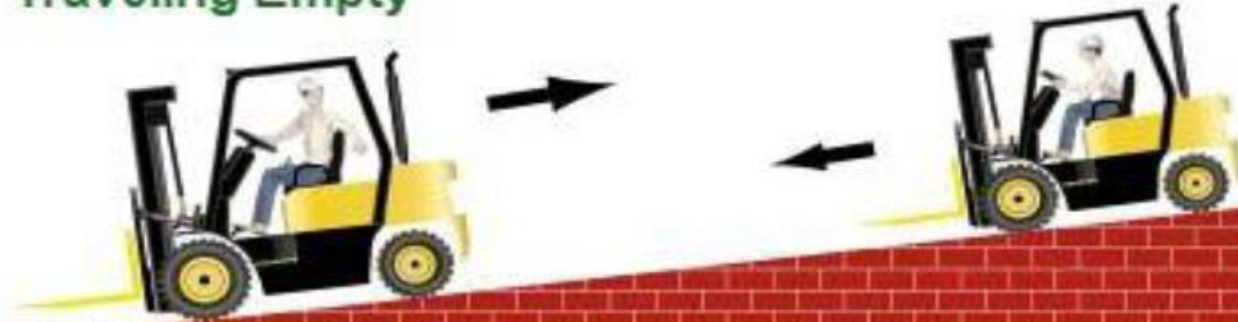
The correct method when descending with the load



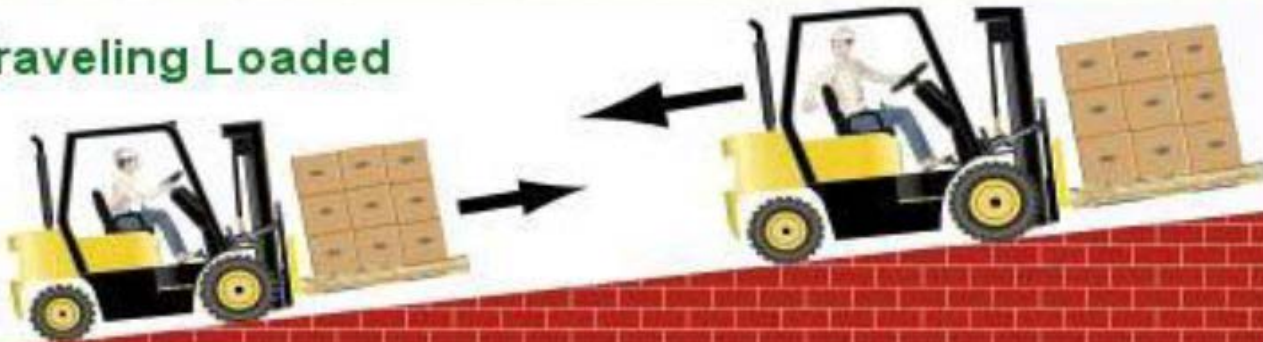


Secure forklift operation

Traveling Empty



Traveling Loaded



Rules for Operating Forklifts on Sloping Roads/Ramps

Always keep the heavy side of the forklift on the high side.

If the forklift is loaded (front side is heavy)

If you want to go up the ramp, you have to drive forward.

When coming down the ramp, drive backwards, that is, the load will always be on the higher side.

If the forklift is not loaded (the rear side is heavy)

If you want to go down the ramp, you have to drive forward.

To get up the ramp, you have to drive backwards.



Safety Rules

- Always wear a seatbelt.
- Never jump off a moving forklift.
- Do not carry passengers in any way.
- Be alert for pedestrians, signal and give way.
- Keep the fork 4-6 inches (10-15 cm) above the ground when running.
- Do not lift any object by tying ropes, cables or chains under the fork.
- Set the fork as wide as possible to carry the load and lock it properly.
- Do not add any extra weight even for a short period of time. Be sure of the weight of the load.





Safety Rules

- Drive the load slightly backwards for stability
- If the load is not visible over the front, run in reverse
- Drive the load about 6 inches above the ground
- Pedestrians always have priority (right of way)
- Remember, the driving wheel is your pivot point
- Comply with traffic rules and local regulations
- Always slow down when making turns

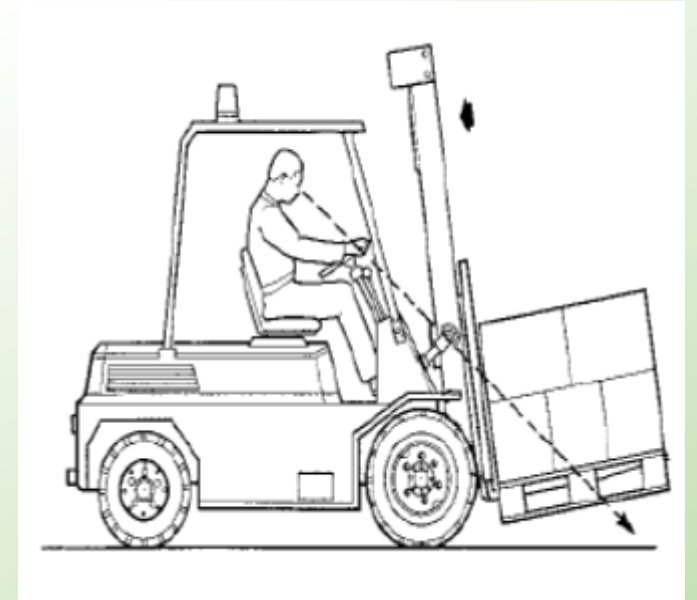




Safety Rules

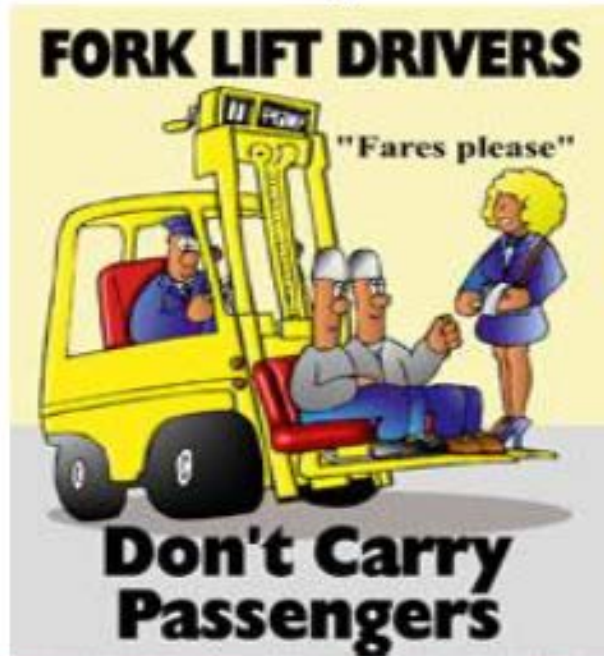
- Tilt the mast slightly backwards before travelling.
- Do not carry loads more than the forklift's designated capacity, not even for a few pounds.
- Do not lift unstable or unbalanced loads.
- Never lift a person without a secure platform.

At normal speeds, the forklift will have a maximum speed of 8 mph (12.8 km/h), but a maximum speed of 3 mph (4.83 km/h) in high-speed areas.





Safety Rules



Thank you for your attention
&
support

